Week 12+ (Optional) — Math Kangaroo-Style Stretch

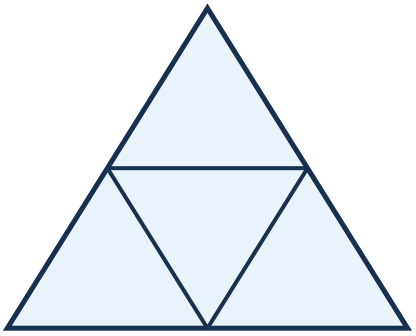
Not part of the core 12-week plan or any Indian olympiad's syllabus — this is the **optional extension** mentioned in `docs/01-syllabus.md` §3 and `docs/03-practice-plan.md`. Attempt it only after you're comfortably scoring well on the 6 sample papers. Math Kangaroo (international) leans harder into spatial reasoning, logic, and "aha" puzzles than SOF-IMO does, so this is a different flavour of difficulty, not more of the same.

15 questions in 3 difficulty tiers, loosely modelled on the real competition's point-value structure (harder questions are worth more). 4-option MCQ throughout, untimed.

Tier 1 — 3 points each (Q1–Q5)

Q1. Look at the pattern: ○ △ ○ △ ○ ? What comes next? A) ○ B) △ C) □ D) ◇

Q2. How many triangles are there in the figure below (count every triangle you can find, including ones made up of smaller ones)?



A) 4 B) 5 C) 6 D) 8

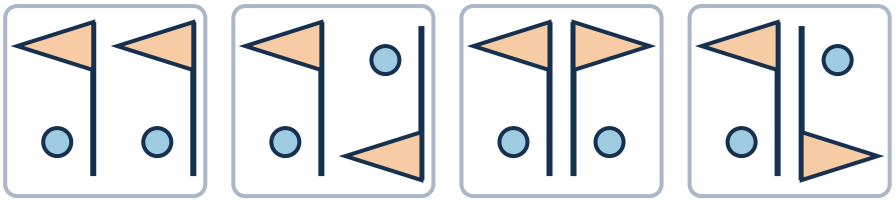
Q3. A bag has 3 red balls and 2 blue balls. Without looking, you take out balls one at a time. What is the smallest number of balls you must take out to be **sure** you have at least one of each colour? A) 2 B) 3 C) 4 D) 5

Q4.

The dashed line is a mirror.



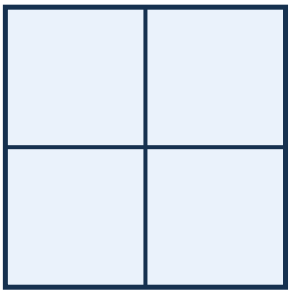
Which option correctly completes the mirror image?



Q5. In a race, Tom finished 2 places ahead of Jerry. Jerry finished 3 places ahead of Spike. If Spike finished 7th, what place did Tom finish? A) 1st B) 2nd C) 3rd D) 5th

Tier 2 — 4 points each (Q6–Q10)

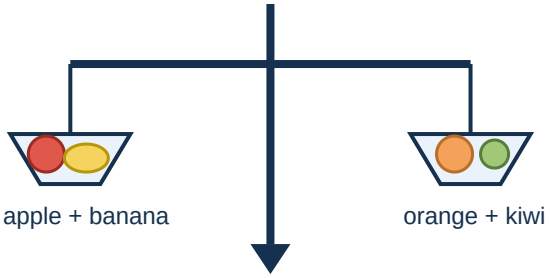
Q6. How many squares are there in the figure below (count every square you can find, including the big one)?



A) 4 B) 5 C) 6 D) 9

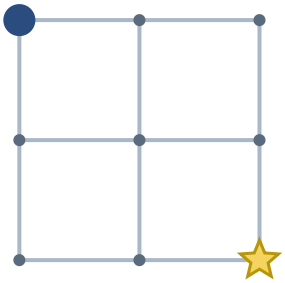
Q7. The sum of two numbers is 50. One number is 4 times the other. Find the smaller number. A) 8 B) 10 C) 12 D) 15

Q8. On the balance scale below, 1 apple + 1 banana balances 1 orange + 1 kiwi, and 1 orange alone balances 2 kiwis. If 1 kiwi weighs 50 g, and the apple and banana weigh the same as each other, how much does 1 apple weigh?



A) 50 g B) 60 g C) 75 g D) 100 g

Q9. Starting at the dot and moving only **right** or **down** along the grid lines, how many different paths lead to the star?



A) 4 B) 6 C) 8 D) 9

Q10. Take a number. Double it, then subtract 7. Multiply that result by 3, and you get 45. What was the original number? A) 9 B) 10 C) 11 D) 13

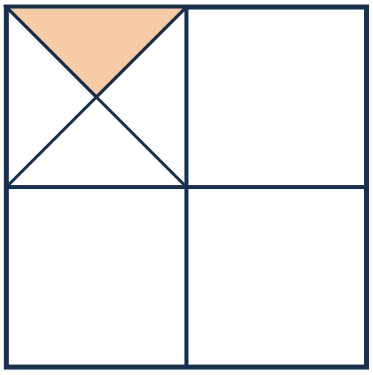
Tier 3 — 5 points each (Q11–Q15)

Q11. Four friends — Ann, Bob, Cara, Dan — each have a different favourite fruit: apple, banana, cherry, date.

- Ann doesn't like banana or date.
- Bob likes cherry.
- Cara doesn't like apple.
- Dan likes date.

What is Ann's favourite fruit? A) Apple B) Banana C) Cherry D) Date

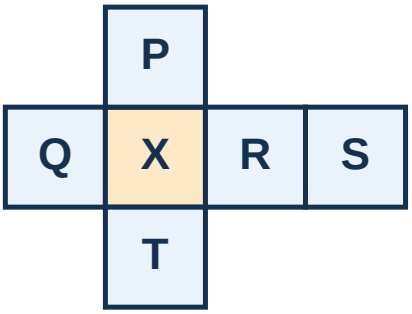
Q12. The large square below is split into 4 equal quadrants. One quadrant is further divided into 4 equal triangles by its two diagonals, and one of those triangles is shaded. What fraction of the **whole large square** is shaded?



A) 1/16 B) 1/8 C) 1/4 D) 1/2

Q13. How many 2-digit numbers have digits that add up to 9? A) 7 B) 8 C) 9 D) 10

Q14. The flat shape below can be folded along its edges to form a cube. Which face will end up **opposite** face X?



A) P B) Q C) R D) S

Q15. A snail is at the bottom of a 10-metre well. Each day it climbs up 3 metres, but each night it slides back down 2 metres (except once it reaches the top, it stops climbing for good and doesn't slide back). On which day does it first reach the top? A) Day 7 B) Day 8 C) Day 9 D) Day 10